M4010104 / M4020104: PROGRAMMING IN PYTHON · DIGITAL UNIVERSITY KERALA · CAPSTONE PROJECT SPECIFICATION

Project 07: Kerala Malayalam News Category Classifier & TF-IDF Vocabulary Profiler

Effort Budget: 10–12 Hours | Core Stack: Pure Python, NumPy, Pandas, Matplotlib, Seaborn, Plotly | Max Marks: 25 | Instructor: Dr. Sunil TT

1. Executive Overview & Domain Context

Kerala's regional digital journalism ecosystem produces tens of thousands of articles daily across topics including Politics, Sports, Business, Cinema, Technology, and Local Administration. Analyzing multi-publisher Malayalam journalism requires managing rich agglutinative vocabularies, compound nominalizations, and varied editorial writing styles across news outlets.

Your team will implement a high-capacity Malayalam NLP news profiler in pure Python that ingests 42,000+ article records, filters unparseable publisher exports, computes document Type-Token Ratios (TTR) and Zipfian distributions, derives a vectorized TF-IDF centroid classifier in NumPy, and benchmarks topic overlap answering the 12 core questions below.

Primary Public Dataset & Telemetry/Data Schema

The project utilizes the verified public Kerala Malayalam News Category Dataset (42k Articles) on Kaggle (**`https://www.kaggle.com/datasets/tolstoyjustin/kerala-malayalam-news-category-dataset`**). The dataset structure captures:

- **Raw Feature Attributes:** **`title`**, **`summary`**, **`body`**, **`category`**, **`source_key`**, **`published_date`**.
- **Engineered / Target Variables:** **`word_count`**, **`char_count`**, **`type_token_ratio`**, **`tfidf_feature_vector`**, **`predicted_category`**, and category centroid cosine similarities.

2. Complete Project Directory & File Architecture

Each team must organize their repository into a standardized, production-grade layout. All application source code must reside inside a dedicated package (**`src/`**), with separate directories for raw/curated data, interactive development notebooks, visual figures, and formal documentation:

```
kerala-malayalam-news-classifier/
├── README.md                          # Repository documentation & GitHub Pages source
├── requirements.txt                   # Dependency specification (numpy, pandas, matplotlib, seaborn, p
├── run_pipeline.py                    # Standalone production CLI application entry point
├── data/
│   ├── raw/
│   │   └── kerala_news_malayalam_raw.csv            # Raw uncleaned input dataset (including simulated/
│   └── processed/
│       ├── clean_kerala_news.csv          # Curated, validated clean CSV dataset
│       └── news_ingestion_audit.log           # Timestamped audit report detailing filtered/rejected re
├── src/
│   ├── __init__.py                    # Python package initialization
│   ├── ingestion.py                   # Defensive line-by-line file reader, range auditor & CSV exporte
│   ├── analyzer.py                    # MalayalamNewsProfiler OOP class, NumPy vectorization & Pandas a
│   └── visualizer.py                  # Matplotlib envelope plots, Seaborn heatmaps & Plotly HTML expor
├── notebooks/
│   └── development_cycle.ipynb        # Development cycle notebook: EDA, trials, tests & unit assertion
└── reports/
    ├── figures/
    │   ├── primary_envelope.png       # Matplotlib confidence envelope & regression overlay
    │   ├── correlation_heatmap.png    # Seaborn annotated feature correlation heatmap
    │   └── distribution_regimes.png   # Seaborn distribution across categorical/domain tiers
```

```
├── interactive_dashboard.html     # Standalone interactive Plotly HTML dashboard with controls
└── Project_Report.pdf             # 3-4 page final technical report
```

3. Dual Development Deliverables: Notebook vs. Standalone Program

To mirror industry software engineering standards, each student group must deliver two distinct, complementary implementations of their solution:

Deliverable A: The Development Cycle Notebook (`notebooks/development_cycle.ipynb`)

The development notebook chronicles your team's engineering journey from raw data exploration to validated model prototypes. It is not merely a final script pasted into cells, but an authentic record of the development cycle:

- **Phase 1: Exploratory Data Ingestion & Diagnostics:** Loading raw data, inspecting missing values, testing boundary rules, and logging rejected records.
- **Phase 2: Mathematical Formulation & Feature Verification:** Developing and validating NumPy array calculations against target benchmarks using `np.allclose()`.
- **Phase 3: OOP Class Architecture & Unit Tests:** Instantiating the `MalayalamNewsProfiler` class, executing automated unit assertions (e.g., verifying that passing invalid parameters raises an explicit `ValueError`), and testing edge cases.
- **Phase 4: Statistical Modeling & Regimen Exploration:** Implementing ordinary least-squares regression, rolling-window calculations, and multi-metric Pandas grouping.
- **Phase 5: Visual Prototype Development:** Iteratively designing and refining Matplotlib and Seaborn figures, followed by configuring the interactive Plotly HTML dashboard.

Deliverable B: The Final Standalone Python Program (`run_pipeline.py` & `src/`)

While the notebook documents the exploratory development cycle, your repository must also provide a production-ready, fully executable Python application. The program must run end-to-end directly from the terminal without requiring Jupyter:

- **Command-Line Interface (CLI):** Executed via `python run_pipeline.py --input data/raw/kerala_news_malayalam_raw.csv --output-dir data/processed --report-dir reports`.
- **Modular Package Import:** The driver script imports cleanly separated domain logic from `src.ingestion`, `src.analyzer`, and `src.visualizer`.
- **Terminal Feedback & Exit Codes:** The program must print structured progress indicators, log clean/faulty record counts to the console, and return an exit code of `0` upon successful pipeline completion.
- **Automated Artifact Generation:** In a single execution pass, the script reads raw inputs, writes `data/processed/clean_kerala_news.csv`, saves `data/processed/news_ingestion_audit.log`, exports static figures to `reports/figures/`, and generates the standalone `reports/interactive_dashboard.html`.

4. Technical Investigation Requirements & Research Questions (Q1–Q12)

Your team's implementation across the development notebook and standalone program must rigorously answer the following 12 course-aligned engineering questions:

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
File I/O & Exceptions (`src/ingestion.py`)	**Q1**	**Article Header & Encoding Integrity Audit:** Parse raw article CSV rows line-by-line with try-except. Validate that required columns (title, category, body) are populated with non-trivial text and handle encoding drops.	open(), try-except (ValueError, UnicodeDecodeError), length checks

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Data Curation & Export (`src/ingestion.py`)	**Q2**	**Curated News Corpus Export & Audit Log:** Write validated articles to clean_kerala_news.csv. Concurrently log empty articles, truncated publisher records, and rejected rows into news_ingestion_audit.log.	Defensive file writing, dictionary parsing, formatted log reporting
OOP Design (`src/ingestion.py`)	**Q3**	**Malayalam News Profiler Class:** Design an encapsulated class MalayalamNewsProfiler with methods load_from_csv(), compute_tfidf_centroids(min_df=10), and summary_report(). Ensure passing min_df < 1 raises ValueError.	OOP class, __init__, encapsulation, raise ValueError
NumPy Arrays (`src/analyzer.py`)	**Q4**	**Vectorized TF-IDF Calculation & Verification:** Using only raw NumPy arrays and dictionary token counters, compute term-frequency and inverse-document-frequency weights for top vocabulary terms. Verify via np.allclose().	np.log(), array broadcasting, np.allclose()
NumPy Masking (`src/analyzer.py`)	**Q5**	**High-IDF Domain-Specific Token Masking:** Identify high-IDF category-discriminative terms in NumPy. Compute Jaccard vocabulary similarity between Politics, Business, and Sports categories using Python set operations.	np.mean(), np.std(), boolean masks (&, |), Python set
NumPy Regression (`src/analyzer.py`)	**Q6**	**Article Body Length vs. Category OLS Model:** Fit an ordinary least-squares model relating article body length to headline word count across news categories via np.linalg.lstsq. Quantify category-specific regression slopes.	np.column_stack(), np.linalg.lstsq(), matrix product (@), R2
Pandas Series (`src/analyzer.py`)	**Q7**	**Publisher Volume & Category Distribution Audit:** Quantify article volume across the 10 publisher sources in Pandas. Identify category imbalance (e.g., Entertainment vs. Business) and compute publisher missingness rates.	.value_counts(normalize=True), .groupby('source_key').count()
Pandas Groupby (`src/analyzer.py`)	**Q8**	**Publisher-by-Category Cross-Tabulation & Pivot Table:** Construct a 2D cross-tabulation of publisher sources vs. top 8 news categories. Build a pivot_table(margins=True) showing mean article word count across source and topic.	pd.crosstab(), .pivot_table(margins=True)
Pandas Reshaping (`src/analyzer.py`)	**Q9**	**Wide-to-Long Category TF-IDF Weight Ranking:** Reshape category-wise centroid TF-IDF vectors into tidy long format (Category, Token, TFIDF_Weight) via pd.melt() and extract the top 5 highest-weighted terms per category.	pd.melt(), .groupby('Category'), .nlargest(5)
Matplotlib (`src/visualizer.py`)	**Q10**	**Matplotlib Zipf's Law Log-Log Frequency Curve:** Plot rank vs. frequency for the top 1,000 Malayalam news tokens on a log-log scale in Matplotlib. Overlay the theoretical Zipf's law line (slope = -1.0) with custom legends.	plt.subplots(), ax.loglog(), ax.plot(), custom legends
Seaborn (`src/visualizer.py`)	**Q11**	**Seaborn Category Vocabulary Overlap Heatmap:** Compute a pairwise Jaccard similarity matrix between all news categories and visualize using sns.heatmap(annot=True). Generate a side-by-side boxplot of word counts by topic.	sns.heatmap(), sns.boxplot(), sns.despine()

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Plotly HTML (`src/visualizer.py`)	**Q12**	**Interactive Category Keyword Sunburst/Explorer:** Build an interactive HTML dashboard (interactive_dashboard.html) displaying news topics, top keywords, and article lengths with interactive search filters.	plotly.express.sunburst() / bar(), hover tooltips, write_html()

5. Collaborative Git Workflow, GitHub Pages & README.md Specification

Best Practices: Collaborative Git Workflow & GitHub Pages Deployment from Day 1

From the very first day, the student team must adopt professional version control and documentation practices. On Day 1, one designated team member initializes the central public GitHub repository, establishes a clear branch protection rule preventing direct pushes to `main`, adds all teammates as registered **Contributors** (with write access), and grants read access to the Course Instructor (`Dr. Sunil TT`) and teaching assistants. Immediately thereafter, the team must activate **GitHub Pages** (via *Settings → Pages → Build and deployment: Deploy from a branch: main /docs or root*) so that project documentation, interactive architecture diagrams, and the live Plotly dashboard (`reports/interactive_dashboard.html`) are publicly accessible and continuously deployed throughout the development lifecycle. All feature development must follow a strict branching model (e.g., `feature/ingestion`, `feature/analysis`, `feature/visualization`): individual team members must write small, atomic, well-commented commits from their own personal GitHub accounts and merge them into `main` exclusively through formal **Pull Requests (PRs)** accompanied by peer review. Dumping entire codebases in a single, monolithic end-of-semester commit is strictly prohibited and will result in zero marks for individual Git contribution; maintaining an unbroken, transparent Git commit and PR audit trail is essential for passing the code ownership defense during the live viva-voce examination.

Repository README.md Specification Template

The root `README.md` serves as the primary technical landing page and must strictly conform to the following structural format:

````
# Project 07: Kerala Malayalam News Category Classifier & TF-IDF Vocabulary Profiler
[![Course](https://img.shields.io/badge/Course-Programming%20in%20Python-blue.svg)](https://duk.ac.in)
[![Python](https://img.shields.io/badge/Python-3.11%2B-brightgreen.svg)](https://python.org)
[![License](https://img.shields.io/badge/License-MIT-yellow.svg)](LICENSE)

## 1. Project Overview & Domain Context
- **Institution:** Digital University Kerala (MTech AI / MSc AI)
- **Course:** M4010104 / M4020104 – Programming in Python | **Instructor:** Dr. Sunil TT
- **Dataset:** Kerala Malayalam News Category Dataset (42k Articles) on Kaggle (https://www.kaggle.com/d

## 2. System Architecture & Pipeline Design
[Provide an ASCII or Mermaid architecture flow diagram showing raw ingestion -> validation -> analyzer -

## 3. Installation & Environment Setup
```bash
git clone https://github.com/<your-team>/kerala-malayalam-news-classifier.git
cd kerala-malayalam-news-classifier
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```

## 4. Execution Instructions
### A. Running the Standalone Application
```bash
python run_pipeline.py --input data/raw/kerala_news_malayalam_raw.csv --output-dir data/processed --repo
```
### B. Running the Development Notebook
```bash
````
````

```
jupyter notebook notebooks/development_cycle.ipynb
```

5. Key Analytical & Statistical Findings (Q1-Q12 Summary)
[Provide a concise markdown summary table detailing raw vs. clean counts, regression coefficients, anoma

6. Interactive Telemetry Dashboard
Explore the live interactive Plotly visualization hosted on GitHub Pages:
🔗 [Live Project Dashboard](https://<your-team>.github.io/kerala-malayalam-news-classifier/interactive_d

7. Contributor Git Commit & Pull Request Log Table
Contributor Name	GitHub Handle	Assigned Module / Tasks	Verified Commit Hash Range	Merged Pull
[Student Full Name]	`@github_handle`	Modular Architecture (`src/ingestion.py`, `src/analyzer.py`,

6. Evaluation Rubric & Mandatory Academic Integrity Governance

Projects are evaluated on a 25-mark scale based on code quality, mathematical correctness, visualization depth, git collaboration discipline, and live technical defense:

Assessment Criterion	Basis of Evaluation & Deliverable Verification	Max Marks
1. Data Curation & File I/O Pipeline (Q1–Q2)	Robust line-by-line file reading with defensive **`try-except`** handling, physical/domain bounds validation, clean CSV output, and structured **`news_ingestion_audit.log`** reporting.	**5 Marks**
2. OOP, NumPy & Pandas Analytics (Q3–Q9)	Modular **`MalayalamNewsProfiler`** class architecture, vectorized NumPy feature re-derivations (verified via **`np.allclose`**), regression modeling, and Pandas rolling/groupby/melt operations.	**6 Marks**
3. Multi-Library Visualizations (Q10–Q12)	Publication-quality Matplotlib confidence envelope/trend plot, Seaborn correlation heatmap and distribution violin plot, and standalone interactive Plotly HTML dashboard.	**5 Marks**
4. Software Packaging, GitHub & Documentation	Clean directory layout, comprehensive **`README.md`**, working GitHub Pages site, **`development_cycle.ipynb`** notebook, standalone **`run_pipeline.py`**, and balanced individual Git commits/PRs.	**4 Marks**
5. Live Presentation & Code Viva-Voce Defense	Technical walkthrough and individual live viva defense where each student must explain or modify lines of code in real time without external assistance.	**5 Marks**

STRICT ACADEMIC INTEGRITY & ZERO-TOLERANCE AI POLICY (0 MARKS PENALTY): All submitted Python source code (**`.py`**, **`.ipynb`**) and written documentation will be screened through an automated **AI-Code & Plagiarism Detection Suite**. If AI-generated code or synthetic documentation is detected at any checkpoint or final submission, **0 Marks will be awarded immediately** for the entire project. Each student must prove authentic individual code ownership during the live viva-voce examination — *Course Instructor: Dr. Sunil TT*.

7. Milestone Schedule, Submission Checklist & Viva Defense Prompts

To ensure systematic progress across the semester, teams must adhere to three project checkpoints and prepare for on-the-spot live code modification challenges during the viva-voce examination:

Milestone Stage	Scheduled Window	Required Deliverables & Verification Milestones
Checkpoint 1 (Data Ingestion)	Week 11 Lab Session	Repository initialized with contributors and GitHub Pages active. **`src/ingestion.py`** operational with line-by-line filtering, **`clean_kerala_news.csv`** exported, and **`news_ingestion_audit.log`** generated.

Milestone Stage	Scheduled Window	Required Deliverables & Verification Milestones
Checkpoint 2 (Analytics & OOP)	Week 13 Lab Session	**`src/analyzer.py`** complete with **`MalayalamNewsProfiler`** class, NumPy feature checks (Q4), outlier masks (Q5), regression modeling (Q6), and exploratory notebook sections.
Final Submission & Viva	Week 15 Project Review	Full package operational: standalone **`run_pipeline.py`**, polished **`development_cycle.ipynb`**, interactive **`interactive_dashboard.html`** on GitHub Pages, 3–4 page PDF report, and live code defense.

Representative Live Viva-Voce Code Modification Challenge

During the individual code viva-voce, the instructor will ask students to execute an immediate 2-minute code modification on their working repository to demonstrate genuine authorship:

- **Challenge Prompt:** *"In src/analyzer.py, adjust the minimum document frequency threshold (min_df) from 10 to 25. Re-run your centroid calculation and report the reduction in vocabulary dimension."*